

Loan Approval Prediction System Using Machine Learning

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United College of Engineering and Research, Prayagraj

BINARY CLASSIFICATION · PREDICTIVE FINANCE

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Problem Statement

Manual Bottlenecks

Manual underwriting introduces throughput constraints, high processing costs, and inconsistent credit decisions.

Type I Error

Approving high-risk borrowers → capital default and financial losses.

Type II Error

Rejecting creditworthy applicants → lost interest income and reduced customer retention.

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Project Objectives

01

Data Cleansing

Handle missing values using mode and median imputation across categorical and continuous features.

02

Exploratory Analysis

Visualize feature distributions, detect outliers, and identify key correlation pathways.

03

Model Optimization

Train and fine-tune ML models to maximize predictive performance and avoid overfitting.

04

Metrics Validation

Evaluate using precision, recall, and F1-scores on an independent test set.

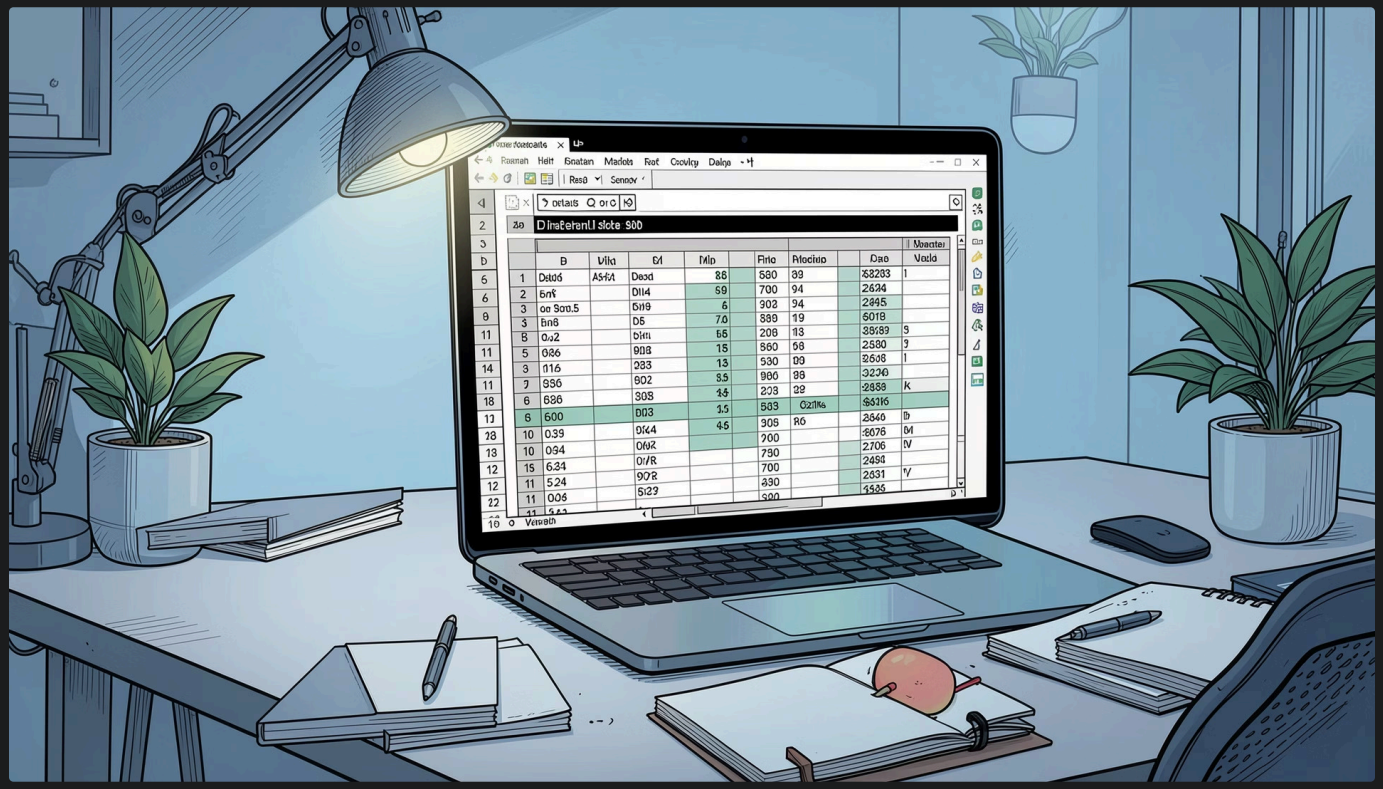
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Web Deployment

Deploy an interactive Flask-based app for real-time automated loan risk assessment.

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Dataset Overview



Kaggle Loan Prediction Dataset

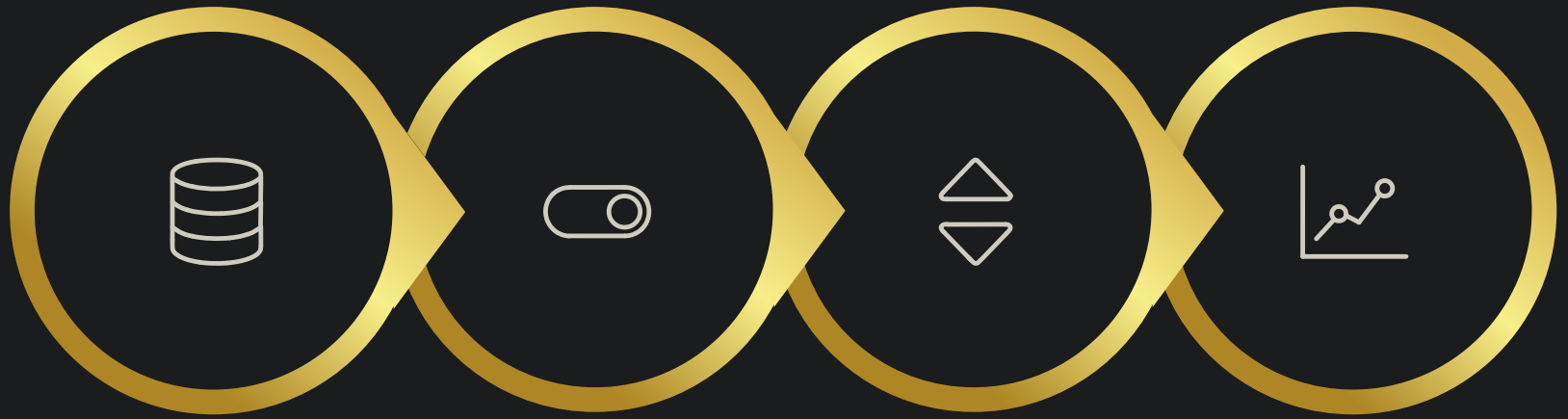
614 borrower profiles · 13 features combining financial and demographic data.

- ApplicantIncome, CoapplicantIncome, LoanAmount
- Credit_History (dominant predictor)
- Property_Area: Urban / Semiurban / Rural
- Target: Loan_Status (1 = Approved, 0 = Rejected)

Class split: 68.7% approved vs 31.3% rejected — imbalanced dataset.

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Data Preprocessing Pipeline



Imputation

Label Encode

Ordinal Map

Log Transform

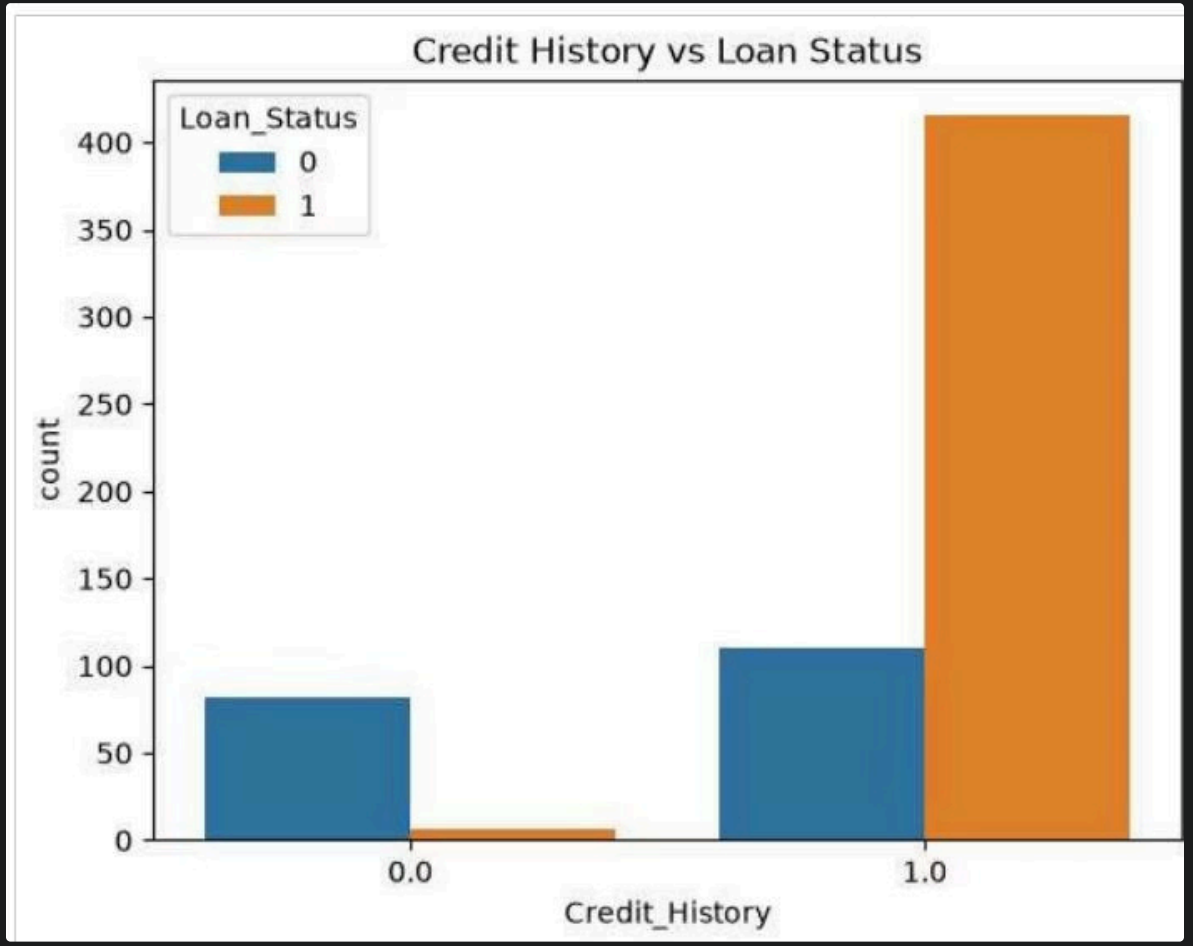
Mode imputation was applied to categorical columns (Gender, Married, Self_Employed, Credit_History). Median imputation handled skewed continuous columns. Log transformation normalized financial distributions, minimizing outlier impact and improving model convergence.

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Exploratory Data Analysis

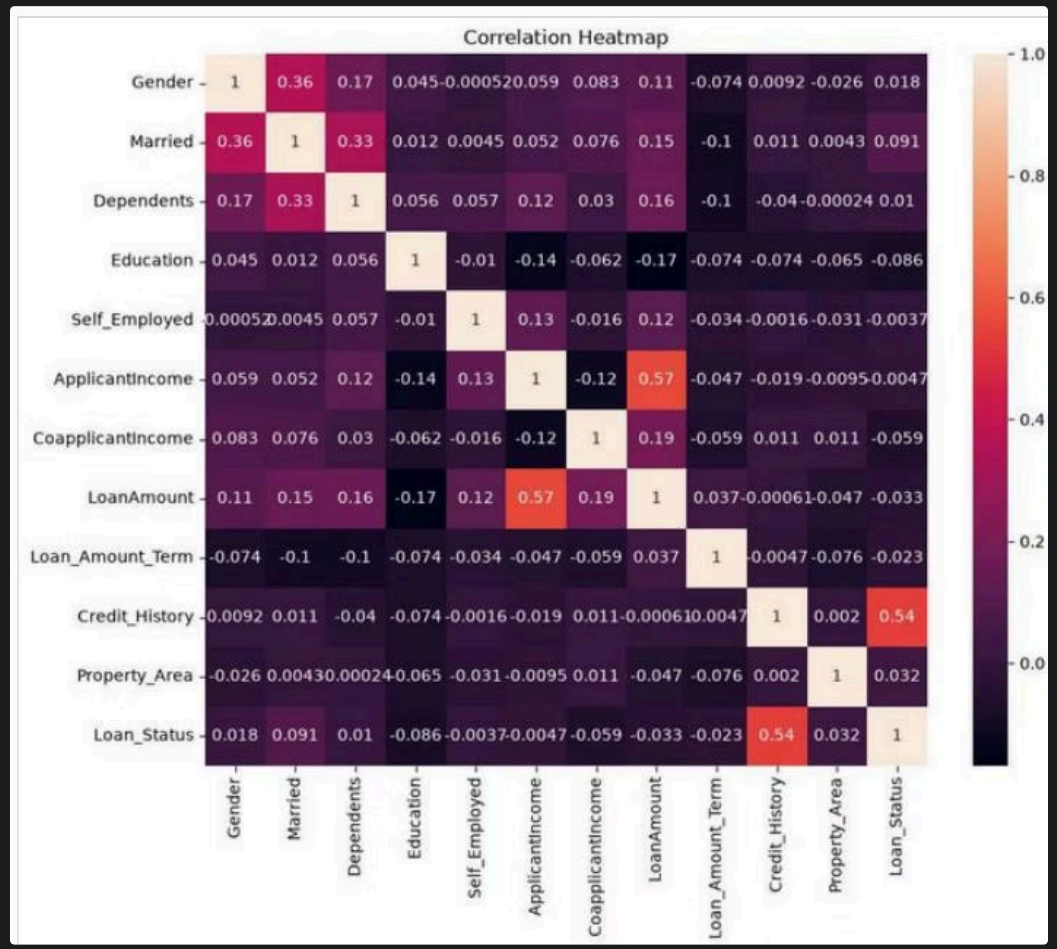
Key EDA Findings

- **Income:** Heavily right-skewed; peak at ₹3,000–5,000 range
- **Loan Amount:** Median 128; outliers extend to 700 units
- **Loan Status:** 422 approved vs 192 rejected (68.7% approval)
- **Credit History:** Strongest predictor — 1.0 score vastly increases approval
- **Correlations:** LoanAmount–Income (0.57); Loan_Status–Credit_History (0.54)



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Correlation Analysis



Pearson Correlation Heatmap

Most feature pairs show near-zero correlation, confirming independent analytical value. Two dominant signals emerge:

- **LoanAmount ↔ ApplicantIncome (0.57):** Loan principals scale directly with applicant revenue.
- **Loan_Status ↔ Credit_History (0.54):** Historical credit compliance is the strongest approval signal.

These findings validate Credit_History as the primary modeling feature.

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Machine Learning Model

Logistic Regression — Selected Model

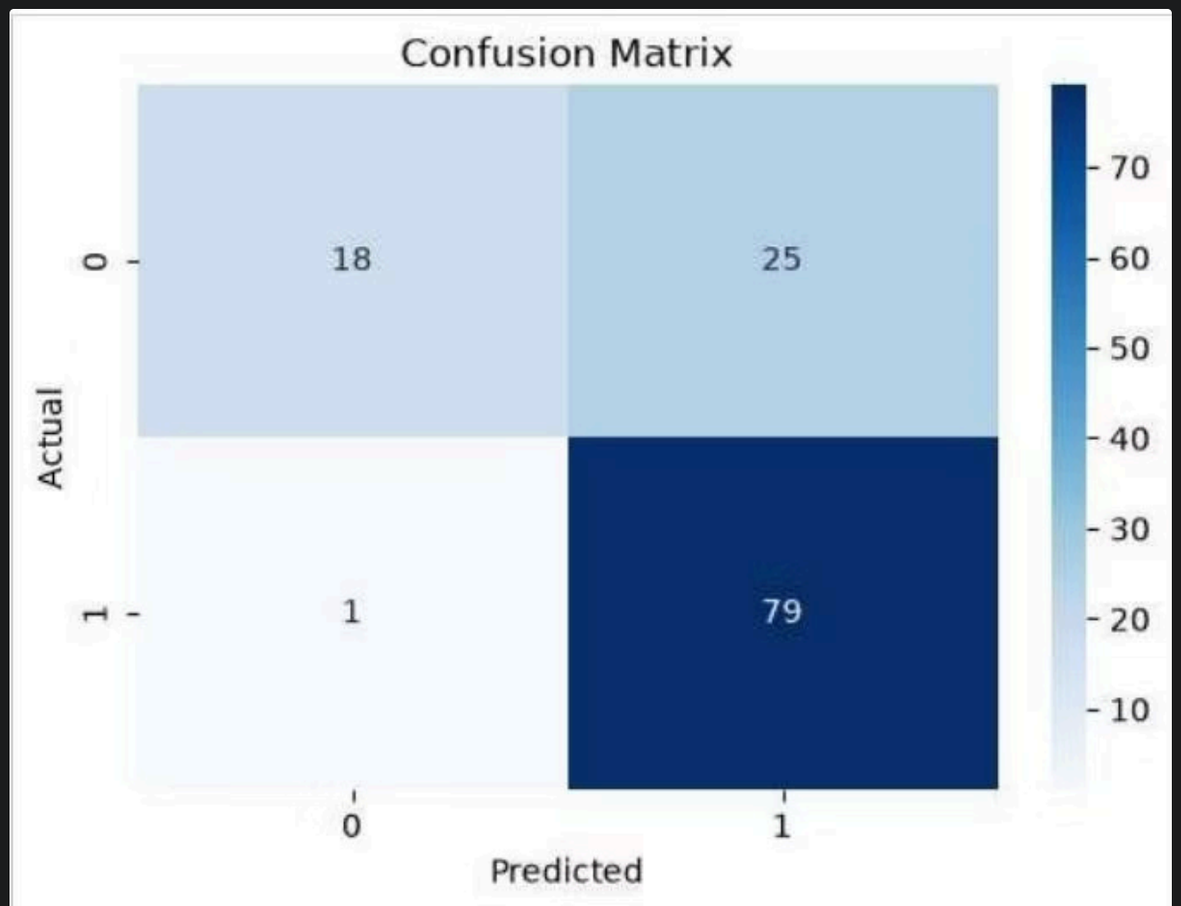
A regularized Logistic Regression model was chosen for its interpretability, stability, and strong baseline performance on tabular credit data.

- Sigmoid function maps inputs to probability [0, 1]
- Log transformation applied to skewed financial features
- Trained on preprocessed Kaggle dataset (614 records)
- Deployed via Flask micro-framework backend



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Model Performance



Test Set Results (123 samples)

TP=79, TN=18, FP=25, FN=1. High recall on approved loans minimizes missed creditworthy applicants.

78.86%	99%	95%	86%
Global Accuracy	Recall (Class 1)	Precision (Class 0)	F1-Score (Class 1)

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Flask Web Application & Conclusion



LoanPredict AI — Deployed Interface

- Dark-themed Flask web app with interactive eligibility form
- Real-time prediction with explicit approval probability (e.g., 83.91%)
- Model: Logistic Regression · Accuracy: 78.86% · 614 applications

Future Scope

- XGBoost / LightGBM ensembles for non-linear patterns
- Alternative credit data (utility payments, mobile transactions)
- SHAP/LIME explainability for regulatory compliance

✓ End-to-end ML pipeline successfully deployed — AIML Internship 2026, IIHMF MNNIT.

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